

Ex. 3

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```
print(uuid::UUIDgenerate())
```

```
## [1] "fb1a28b5-5d91-4a4a-bad1-ff8950e462a7"
```

Question 1

```
rent <- c(12500, 13800, 11200, 15000, 12500, 26000,  
         14300, 13100, 12500, 16700, 14900, 13600)
```

```
mean(rent)
```

```
## [1] 14675
```

```
median(rent)
```

```
## [1] 13700
```

```
table(rent)
```

```
## rent  
## 11200 12500 13100 13600 13800 14300 14900 15000 16700 26000  
##      1      3      1      1      1      1      1      1      1      1
```

The mean rent is 14,675 NTD, the median rent is 13,700 NTD, and the mode is 12,500 NTD because it appears most frequently (3 times).

Question 2

```
rent_cleaned <- rent[rent != 26000]  
mean(rent_cleaned)
```

```
## [1] 13645.45
```

```
median(rent_cleaned)
```

```
## [1] 13600
```

1. The new mean is approximately 13,645.45 NTD and the new median is 13,600 NTD.
2. The mean changes much more (a decrease of over 1,000 NTD) compared to the median (which only decreases by 100 NTD).
3. The median better describes the typical rent near campus because it is resistant to extreme values (outliers), whereas the mean is pulled upward by unusually expensive apartments.

Question 3

```
weights <- c(0.36, 0.32, 0.32)
scores  <- c(88, 72, 80)

final_grade <- sum(weights * scores) / sum(weights)
final_grade
```

```
## [1] 80.32
```

The weighted mean is 80.32, so the student's final grade is 80.32.